ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Robotics and Industrial Automation paper 1 · practice a.

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How to mark: each ✓ is one mark. Accept any standard ladder notation and addressing. Answers must relate to the case study where asked. Accept other correct answers.

SECTION A

QUESTION 1

1.1 1.1.1 D ✓ 1.1.2 B ✓ 1.1.3 D ✓ 1.1.4 B ✓ 1.1.5 C ✓ 1.1.6 A ✓ 1.1.7 B ✓ 1.1.8 D ✓ 1.1.9 A ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 False ✓ 1.2.4 True ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 A ✓ 1.3.2 D ✓ 1.3.3 F ✓ 1.3.4 E ✓ 1.3.5 I ✓ 1.3.6 J ✓ 1.3.7 G ✓ 1.3.8 C ✓ 1.3.9 H ✓ 1.3.10 B ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Any FIVE, name ✓ + function ✓: CPU – executes the program; power supply – provides power (usually 24 V DC) to the PLC; input module – receives signals from sensors and switches; output module – switches motors, valves and lamps; memory – stores the program and data; programming device / software – used to write and download the program; communication module – connects to HMI, robot, network (10)

2.2 Any FOUR: easy to change the program without rewiring ✓; fewer wires and less panel space ✓; faster fault-finding (monitoring, diagnostics) ✓; reliable – no mechanical relay wear ✓; timers and counters built in; can communicate with HMIs and SCADA; cheaper for complex systems (4)

2.3 (a) input, digital ✓ (b) input, analogue ✓ (c) output, digital ✓ (d) output, analogue ✓ (4)

[18]

QUESTION 3

3.1.1 It is a latching (seal-in / holding) contact ✓ – it keeps the motor running after START is released ✓. (2)

3.1.2 Pressing START energises Q0.0 and the conveyor motor starts ✓; Q0.0 latches ✓; the Q0.0 contact in rung 2 starts timer T1 ✓; after 10 s, T1's contact closes and the warning lamp Q0.1 switches on ✓. (4)

3.1.3 The motor stops (latch is broken) ✓; T1 resets so the lamp switches off ✓. (2)

3.1.4 Fail-safe: if the wire breaks or the contact fails, the input goes OFF and the motor stops ✓, so a broken stop circuit cannot prevent stopping ✓. (2)

3.2 Rung 1: I0.3 (NO) → CTU C1 count input with PV = 24 ✓✓; I0.4 (NO) → reset input of C1 ✓✓; rung 2: C1 contact (NO) → Q0.2 coil ✓✓ (6)

3.3 START buttons I1.0 and I1.1 in parallel ✓✓, with Q0.3 latching contact also in parallel with them ✓; STOP inputs I1.2 and I1.3 (NO contacts in the program, since the buttons are NC in the field) in series ✓✓; Q0.3 coil ✓ (6)

[22]

QUESTION 4

4.1 Any FIVE with function ✓✓✓✓✓: six-axis robot – picks and places packets; robot controller and teach pendant – runs and programs the robot; end effector (vacuum gripper) – holds the packets; conveyor – brings packets; sensors (photo-eye / vision) – detect packet position; PLC – controls the conveyor and coordinates the cell; safety fence, interlocked door, light curtain – protect people; HMI – operator control (5)

4.2 Any FOUR, each described ✓✓: interlocked guard door that stops the robot when opened; robot switched to manual / reduced-speed mode with an enabling (dead-man) switch on the pendant; lock-out / tag-out of power and air before clearing jams; emergency stop buttons inside and outside the cell; light curtains / safety mats at openings; trained and authorised operators only; risk assessment and safe work procedure (8)

4.3 The PLC sets an output "packet ready" when the sensor detects a packet in position ✓; the robot controller reads this signal and starts picking, then sends "pick complete" back to the PLC ✓; the PLC then moves the conveyor to the next packet – each step waits for the other's confirmation ✓. (3)

[16]

QUESTION 5

5.1 An HMI is a local operator panel (touch screen) for one machine or cell ✓✓; SCADA is software on a central computer that supervises, controls and records data from many machines, PLCs or sites ✓✓. (4)

5.2 Any FIVE ✓✓✓✓✓: running / stopped status; number of packets / boxes completed; conveyor speed; alarms and faults (e.g. gripper vacuum lost, door open); emergency stop status; production rate per hour; robot mode (auto/manual); start, stop and reset buttons (5)

5.3 Any THREE: sensors and data analysis predict breakdowns (predictive maintenance) ✓; real-time production data helps managers improve efficiency ✓; machine vision checks quality automatically ✓; remote monitoring; digital twin to test changes; automatic ordering of materials (3)

5.4 Any TWO risks with a solution (½ + ½ each): hacking / cyber-attacks – firewalls, network segmentation ✓; data theft – encryption, strong passwords; malware stopping production – updates, backups, antivirus ✓ (2)

[14]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE